How Wash Traders Exploit Market Conditions in Cryptocurrency Markets*

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Abstract

Wash trading, the practice of simultaneously placing buy and sell orders for the same asset to inflate trading volume, has been prevalent in cryptocurrency markets. This paper investigates whether wash traders in Bitcoin act deliberately to exploit market conditions and identifies the characteristics of such manipulative behavior. Using a unique dataset of 18 million transactions from Mt. Gox, once the largest Bitcoin exchange, I find that wash trading intensifies when legitimate trading volume is low and diminishes when it is high, indicating strategic timing to maximize impact in less liquid markets. The activity also exhibits spillover effects across platforms and decreases when trading volumes in other asset classes like stocks or gold rise, suggesting sensitivity to broader market dynamics. Additionally, wash traders exploit periods of heightened media attention and online rumors to amplify their influence, causing rapid but short-lived spikes in legitimate trading volume. Using an exogenous demand shock associated with illicit online marketplaces, I find that wash trading responds to contemporaneous events affecting Bitcoin demand. These results advance the understanding of manipulative practices in digital currency markets and have significant implications for regulators aiming to detect and prevent wash trading.

Keywords: Wash Trading, Cryptocurrencies, Bitcoin, Market Manipulation, High-Frequency Trading, Trading Volume

JEL Classification: G12, G13, G18, G23, D82, K22

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1 Introduction

Wash trading is the act of simultaneously placing buy and sell orders for the same asset to generate artificial volume. It has long been prohibited in regulated financial markets under provisions such as Section 17(a) of the U.S. Securities Act of 1933. Despite clear rules and surveillance infrastructure in equities and commodities markets [CFTC, 2024], enforcement in cryptocurrency markets remains challenging. The anonymous nature of digital assets, fragmented exchange structures, and inconsistent regulatory classifications have allowed wash trading to persist at scale. Recent research suggest that a substantial portion of trading volume on some crypto exchanges may be non-genuine [Cong et al., 2023]. This paper investigates whether wash trades in Bitcoin are strategically timed to exploit specific market conditions.

Recent enforcement actions by the U.S. Securities and Exchange Commission (SEC) highlight the need to better characterize such behavior. In SEC v. Sun (2023), the agency tracked suspicious transactions across affiliated accounts and used these patterns to allege systematic wash trading [1] [2] The case highlights an emerging challenge: not all wash trades are alike, and their legal treatment increasingly depends on demonstrating intent or strategic manipulation. In response, academic research has begun to move beyond detection toward identifying behavioral patterns, linking wash trades to incentives, information environments, and broader market dynamics.

Using transaction-level data from Mt. Gox, the largest Bitcoin exchange by volume during its peak, I examine whether wash trading responds systematically to liquidity conditions, media cycles, and external demand shocks. The Mt. Gox dataset provides rare access to internal identifiers, allowing for credible identification of self-matched trades. I test whether wash trades intensify during periods of low legitimate trading, decline when traditional assets like gold or equities are more active, and spike during moments

[1] SEC took action against crypto entrepreneur Justin Sun and associates for fraudulently manipulating the secondary market for crypto asset securities Tronix (TRX) through extensive wash trading. The case also involved notable celebrities such as Lindsay Lohan, Jake Paul and others.

[2] In another case, SEC filed 13 charges against Binance Founder and one of the charges alleges that Sigma Chain, an entity controlled by Binance Founder Changpeng Zhao engaged in wash trading to inflate the platform's trading volume.

of heightened investor attention or anticipated Bitcoin demand. The results suggest that wash trades are not random but strategically timed, with implications for regulation, market integrity, and surveillance design in digital asset markets.

This paper contributes to three strands of literature. First, theoretical models of manipulation show how uninformed traders may mimic informed ones to influence prices and extract rents [Glosten and Milgrom, 1985, Kyle, 1985, Allen and Gale, 1992]. Second, recent legal and institutional work highlights the difficulty in prosecuting manipulation without empirical tools to demonstrate intent [Fox et al., 2018, Kyle and Viswanathan, 2008]. Third, emerging empirical studies have documented widespread wash trading in crypto markets [Gandal et al., 2018, Aloosh and Li, 2024], but few have examined its strategic timing. This paper fills that gap by providing empirical evidence that wash trading intensity responds systematically to variations in legitimate trading volume, investor attention, cross-asset liquidity, and demand shocks.

2 Background and Literature Review

In early 2014, a data leak from Mt. Gox provided access to approximately 18 million buy and sell transactions from April 2011 to November 2013. This dataset differs from proprietary exchange data or Bitcoin blockchain transactions because it consists of back-end files recorded by the exchange itself. While blockchain transactions are publicly available, they often do not account for fiat-to-crypto transactions. Many exchanges also do not record internal trades on the public blockchain, instead managing trades among internal wallets and ensuring adequate crypto assets for deposits and withdrawals. Proprietary data released by exchanges may be filtered, potentially omitting undesirable transactions before sharing with third-party providers. Notably, the leaked Mt. Gox data includes a separate file for April 2013, which is speculated to contain altered transactions intended for auditors and investigators.

The data includes transaction ID, amount, time, currency, transaction fees and other peripheral information. Each Mt. Gox user has a unique ID number, allowing transac-

tions to be linked to specific actors. I supplement this dataset with additional variables from various sources. A list of these variables is provided in Table 1.

Characterizing cryptocurrency wash trading is a growing area of research. Gandal et al. [2018] find evidence of wash trading in the Mt. Gox dataset, identifying two bots—"Markus" and "Willy"—that executed suspicious wash trades from 2013 onwards. Aloosh and Li [2024] identify wash trading from June 2011 to May 2013 characterized by the same buyer and seller trade IDs. Chen et al. [2019] use the ratio of a digital currency exchange's media popularity to its off-chain transactions as a proxy to estimate the severity of wash trading. Cong et al. [2023] identify evidence of wash-trading using proprietary data from several exchanges through novel statistical methods.

This article correlates strategic factors with verified wash trade activity. While empirical methods can identify wash trading using statistical techniques, pinpointing wash trading activity is challenging. Whether bad actors engage in strategic wash trading remains an open question that could enhance current methods of identifying wash trades.

2.1 Cryptocurrency wash trading is different from traditional wash trading

In models of trade manipulation [Allen and Gale, 1992, Aggarwal and Wu, 2006], a pooling equilibrium exists between informed traders and manipulative traders. Manipulators execute buy orders to drive prices up and then sell off, often spreading rumors to amplify the effect—a significant portion of which occurs online. Aggarwal and Wu (2006) review SEC cases and find that 54% of manipulators both buy and sell stocks to profit, while about 13% corner the supply to inflate prices. They also find that these manipulations often involve penny stocks, which require less capital and have noisier information environments.

While the market for traditional assets is well-studied in the trade manipulation literature, relatively less is known about cryptocurrencies, which do not represent any claims to fiat currency or ownership, and do not represent a tangible hold on physical goods. Gandal et al. [2018] find that 843 different cryptocurrencies with large volume jumps due

to manipulative activities have their prices increased significantly. This article uses the Mt. Gox dataset to explore strategic factors affecting the dynamics of wash trades.

2.2 Strategic wash trading against on-chain transactions, market transactions and other assets

From earlier, theory suggests that wash traders disguise as informed traders and thus, manipulate other investors into buying the cryptocurrency. As Bitcoin is also a speculative tool, wash traders may observe price movements in traditional commodities like gold, which is often considered as a hedge against inflation, to adjust their strategies. By monitoring correlations between assets, traders may artificially inflate cryptocurrency trading volumes to create misleading market activity based on perceived price relationships.

For instance, if gold prices are rising, a wash trader might simultaneously buy and sell Bitcoin to mimic this trend, creating the illusion of increased investor interest in both assets. This exploits behavioral biases, leading other market participants to perceive the cryptocurrency's price movement as a response to gold's trend.

In this paper, I use high-frequency data of assets to study if there is a correlation between these two factors. High-frequency data captures price movements at a minute-by-minute level and allows me to detect anomalies that may be invisible in lower-frequency data. This granularity enables the identification of the exact timing and extent of wash trades.

H1: Bitcoin's wash trading on one exchange leads trading in other platforms

H2: Bitcoin's wash trading decreases when liquidity in other asset classes increase

2.3 Strategic wash trading through spreading rumors and dark web illicit activities

Aggarwal and Wu [2006] also show that the spread of rumors can explain why uninformed, noisy traders converge on the volume surge and cause the cryptocurrency to rally. In 2011 to 2013, Bitcoin was nascent and Bitcoin trade mainly comprised two uses. Firstly,

Bitcoin was used as a currency to perform transactions on the dark web [Christin, 2013, Foley et al., 2019]. Many studies show that Bitcoin was favored as a transaction platform for illicit activities due to its anonymous nature and lack of regulation. As much as 76 billion dollars of illegal activity per year involve bitcoin in later years, while Christin [2013] documents that almost 20% of all Bitcoin transactions were through the Silk Road, a large dark web marketplace for goods and services.

Second, Bitcoin served as a speculative instrument. The speculative nature of Bitcoin is also well-documented, with the meteoric rise in price from $0.01 in 2010 to a high of $73,737 in March 2024. Schilling and Uhlig [2019] show that a two currency economy with fiat and bitcoin can reach equilibrium with bitcoin speculation. Both speculation and currency usage drive the price of bitcoin. As such, press and Internet coverage and rumors could drive speculation, which affects the price of cryptocurrencies. Wash traders could then potentially use this channel to enhance their wash trades. Thus, I test whether a weaker information environment is exploited by wash traders to strategically increase their volume of wash trades.

The Dark Web Silk Road was an online black market, notorious for facilitating the anonymous trade of illegal goods and services. Launched in 2011 by Ross Ulbricht, the platform used the anonymity of the Tor network and Bitcoin transactions to evade law enforcement. At its peak, Silk Road became a haven for illicit activities, predominantly drug trafficking, but also including weapons sales, counterfeit documents, and hacking services. It disrupted traditional drug distribution channels, introduced sophisticated encryption and payment systems, and caused regulators difficulties in policing cyberspace. Silk Road's eventual shutdown in 2013 by the FBI. Because Silk Road transactions were heavily financed by Bitcoin, events on Silk Road affected Bitcoin's demand. One such event is Pot Day on April 20th, 2012 [Christin, 2013]. I exploit the exogenous shock to show causal evidence that wash trades are strategically increased in the time period leading up to Pot Day.

I state my next two hypotheses in the alternate form below.

H3: Bitcoin's wash trading manipulative effect is higher when media attention is high

H4: Bitcoin's wash trading increases during exogenous demand events

3 Data and Results

The Mt. Gox dataset consists of two folders: *backoffice* and *trades*. The *backoffice* folder contains peripheral information related to the leak, while the *trades* folder includes the leaked transaction-level trade files. I follow the standard de-duplication procedures used in prior literature, as documented in Gandal et al. (2018) and Aloosh and Li (2024). Further details are provided in Appendix A1. The extracted variables, along with other variables used in the analysis, are described in Table **??**.

Using the variables listed in Table **??**, I present descriptive statistics for selected key variables in Table **??**.

3.1 Do Crypto Wash Trades' Volume or Value Matter?

To assess whether the volume of BTC wash trades matters, I first estimate the total circulating supply of Bitcoin in the market. New Bitcoins are minted daily, with the total supply expected to cap at 21 million BTC by the year 2140. Market capitalization is a critical metric, particularly because it has been cited in legal defenses. For instance, in v. Sun [2024], defense attorneys argued that wash trading accounted for only a negligible fraction of the total market capitalization, using this to contest the materiality of the charges.

I collect data on the total circulating supply of Bitcoin from *bitcoinexplorer.com* on several dates and apply linear interpolation to estimate daily supply figures. In untabulated results, I find that the average wash trade accounts for approximately $6.5 \times 10^{-5}\%$ of the total market capitalization. This is considerably lower than the 0.0039% figure cited in the aforementioned SEC case. Therefore, I conclude that the capitalization of documented wash trades in the Mt. Gox data is economically insignificant.

Next, to determine how wash-traders decide when to wash-trade, I first turn to Classification and Regression Trees (CART) models. CART offers a non-parametric approach

to predict outcomes based on a series of decision rules derived from the MtGox data. By constructing a binary decision tree, CART models segment the data into subsets that are increasingly homogeneous with respect to the occurrence of wash trades. This segmentation helps to identify critical thresholds and patterns in the data that influence wash trading behavior.

CART also excels in handling complex, non-linear relationships within the data, making it well-suited for financial markets where interactions between variables can be intricate and unpredictable. Consequently, this helps in developing strategies to monitor and potentially mitigate such manipulative trading activities.

I use four popular methods of CART analysis to perform my tests. I use 1) standard CART algorithm, 2) random forest, an ensemble learning method that uses multiple decision trees to improve model accuracy and reduce overfitting 3) gradient boosted tree, where decision trees are built sequentially and each tree corrects the errors of the previous one. I implement XGBoost (Extreme Gradient Boosting) for this, which incorporates advanced features such as regularization to prevent overfitting, parallel processing for faster computation, and handling of missing values. Lastly, I use 4) AdaBoost, an ensemble learning method that focuses on improving the accuracy of weak classifiers by iteratively adjusting the weights of misclassified data points.

I use five variables in my CART analysis:

1. Non-wash trade volume (in 30-minute intervals),
2. Wash trade volume (in 30-minute intervals),
3. Total trade volume (in 30-minute intervals),
4. Liquidity of the BTC market on Mt. Gox, measured using the Amihud illiquidity measure, and
5. Realized volatility of BTC on Mt. Gox.

Before performing CART, I verify that all five time series are stationary and free of time trends using the Augmented Dickey-Fuller (ADF) test. In my CART models, I

adopt a train-test split ratio of 70:30, a maximum tree depth of 3, and designate next period's wash trade volume as the target variable.

In Table **??**, I report model results using lagged values at $t-1$, $t-2$, and $t-24$. In extended analysis presented in the Appendix, I include additional lags: $t-1$, $t-2$, $t-3$, $t-4$, $t-5$, $t-6$, and $t-24$.

For robustness, a random placebo variable is added to each model. Any explanatory variable with lower importance than the placebo is disregarded in interpretation for that specification.

To strengthen my analysis, I include two other deep learning models. Firstly, I use Gated Recurrent Units (GRU), a type of recurrent neural network (RNN), to predict the extent of wash trading. GRUs are specifically engineered to manage sequential data by utilizing gating mechanisms that adeptly retain long-term dependencies and control information flow through the network. This feature makes GRUs exceptionally effective in processing time-series data.

Secondly, I use Long Short-Term Memory (LSTM) networks, a sophisticated variant of recurrent neural networks (RNNs). LSTMs are designed to address the vanishing gradient problem, which often plagues traditional RNNs. This problem affects the network's ability to learn and remember long-term dependencies in sequential data. The architecture of LSTM networks includes a unique design that enables them to retain information over extended time periods, making them particularly well-suited for time-series prediction and sequence modeling tasks. For both the deep-learning models, I use the same train-test ratio of 7:3 and 20 epochs with batch size of 32.

A visualization of Table 2 is provided in Appendix A6. As total trade volume may not be robust from wash trade volume, I prioritize the other variables. From Table 3 Columns (3),(4), I can determine that volatility and non-wash volume affects wash-trades, immediately after total trade volume. They are also more important than the placebo, which provides evidence that under decision models, wash traders consider the volume of non-wash trade before making wash trades. In columns (5),(6), the deep learning models instead place the placebo very low in importance, instead showing that non-wash volume

in $t-2$ and $t-3$ affect the decision to do wash-trading the most. Again, this confirms my hypothesis that wash trades are strategic in nature and are based on volume of non-wash trades.

Next, to determine the conditions that are correlated with wash-trading, I next perform a vector autoregression (VAR) analysis framework. I consider the VAR model below

I estimate the following vector autoregression (VAR) model:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t,$$

where y_t is a 5×1 vector of endogenous variables observed at 30-minute intervals. The vector includes: (1) wash trading volume (in BTC units), (2) non-wash trading volume (in BTC units), (3) the exchange rate of BTC to USD (per BTC), (4) the Amihud illiquidity measure, and (5) the realized volatility of Bitcoin returns. Prior to estimation, I confirm that each time series is stationary and free of deterministic time trends by rejecting the null hypothesis of a unit root using the Augmented Dickey-Fuller (ADF) test.

For robustness, I first conduct a Granger Causality test between the variables. The Granger Causality test helps to determine whether one time series can predict another time series. The results show that wash trading volume Granger-causes non-wash trading volume, the BTC to USD exchange rate, the Amihud liquidity measure, and the realized volatility of Bitcoin returns. This causality suggests that fluctuations in wash trading volume can be used to predict future movements in these variables.

Next, I conduct a Johansen cointegration test that identifies the presence of long-term equilibrium relationships between the multiple time series variables. To do this, I estimate a vector error correction model (VECM) derived from the VAR model and the test evaluates the null hypothesis of no cointegration against the alternative of cointegration among the variables. These results are presented in Table 4.

Lastly, I perform a cross-correlation analysis. Cross-correlation measures the similarity between two time series as a function of the time-lag applied to one of them, which helps identify leading and lagging relationships between the variables. The cross-correlation functions (CCFs) are computed for each pair of the five variables over multiple lags

and indicates whether changes in one variable are systematically followed by changes in another.

From Table 4, Panel A, the results of the Johansen cointegration test indicate the presence of at least one cointegrating relationship at the 5% significance level. This finding suggests that there exists a stable, long-term equilibrium relationship among the wash trading volume, non-wash trading volume, total volume, the Amihud measure of liquidity, and the realized volatility of Bitcoin returns. The existence of cointegration implies that despite short-term deviations, the variables tend to move together in the long run.

Panel B shows that non-wash trading activity and volatility may be driving wash trade activity. Liquidity does not appear to be significant, consistent with the results reported in Table **??**. In Panel C, Column (2), we observe that when non-wash trades increase at time t, the coefficients at horizons $t+1$ and $t+2$ are negative. This suggests that wash trade activity decreases in the periods following an increase in non-wash trading. These results provide further evidence that wash trades decline in response to rising legitimate trading activity.

3.2 Can On-Chain and Cross-Market Data Predict Wash-Trading?

Wash traders may base their illicit activity on transactions in other exchanges and on-chain transactions. In Chen et al. [2022], the authors use the ratio between an exchange's percentage of overall market transactions and its percentage of overall media popularity as a gauge of potentially suspicious activity. Their method depends on the availability of data across all exchanges and the calculation of relative ratios to infer the presence of wash trading. It also assumes that all transactions must go through publicly listed hot and cold wallet addresses revealed through Proof-of-Funds disclosures by the exchange. However, not all wallet addresses must be disclosed—and they typically are not—for reasons of security and privacy. Exchanges generally reveal only a subset of wallets to demonstrate solvency, not an exhaustive disclosure of all assets. Furthermore, Mt. Gox's

wallet addresses were not publicly available during the sample period.[3]

Thus, in this section, I examine whether volume in other public exchanges and on-chain activity can serve as leading or lagging indicators of wash trading. My underlying argument is that if legitimate transactions are driven by insider information, on-chain and other exchange transactions should mirror those of the target exchange. Similarly, wash trades in one exchange may spark legitimate trading activity elsewhere. Understanding the extent and pattern of these relationships is essential to characterizing the dynamics of wash trades.

I first identify on-chain transactions from Bitcoin block 145,500 to block 182,600. On-chain transactions are publicly verifiable records stored across Bitcoin nodes worldwide. There are 9,267,803 transactions across these blocks. These records are immutable once created, forming one of the core features of the blockchain: all past transactions are permanently and transparently recorded. The blocks span from September 16, 2011 to June 1, 2012.[4]

A snippet of the data format is shown in Appendix A3. I aggregate the transaction volume into 30-minute intervals to match the granularity of the Mt. Gox dataset. I then divide the Mt. Gox dataset into four quartiles based on daily wash trade volume, computed from 48 half-hour segments per day. The dataset spans 259 days, and I generate a visualization showing the quartile classification of each day.

Next, I conduct a linear regression of $onchain_t$ on $nonwash_t$, separated by wash trade quartiles, where t represents 30-minute intervals. Additionally, I perform a two-step Engle-Granger cointegration test between $onchain_t$ and $nonwash_t$. The optimal number of lags is determined using the Akaike Information Criterion (AIC). The results of the regression and cointegration analysis are presented in Table 3, Panel A and Panel B, respectively.

Next, I also identify a list of 30 other exchanges that linked their APIs to *bitcoin-*

[3] Arkham Intelligence, the most comprehensive major cryptocurrency analytics provider, only provides data on Mt. Gox's publicly listed wallet addresses starting from 2014.

[4] The data were retrieved via continuous API calls to `blockchain.info`. Due to timeout and network errors, some blocks may be missing. Gathering this data took approximately one week of uninterrupted API requests.

charts.com, building a comprehensive, daily dataset of total BTC (USD) market activity from 26 June 2011 to 20 May 2013. *Bitcoincharts.com* is a resource commonly used to measure total BTC market activity during this time period, and the data is considered high quality. In Appendix A3, I plot a distribution of Mt. Gox activity against the total BTC-USD market activity over the same timeframe. I achieve a mean of 83%, which is comparable to past literature. Many exchanges during this time were short-lived, and the completeness of total market activity records may lead to slight discrepancies in Mt. Gox's share of BTC transaction volume.

Since the market activity volume is reported at a daily frequency, I use a regression approach rather than high-frequency methods, as I am unable to identify intra-day time trends, unlike with the on-chain data. I present the linear regression of $onchain_t$ on $nonwash_t$, split across the four quartiles of wash trading volume, where t represents 30-minute intervals, in Table 3, Panel C.

Table 5, Panel A, indicates that in an OLS regression, Quartile 3 non-wash trade volume correlates most closely with on-chain BTC volume. In Table 5, Panel B, a lower p-value suggests stronger evidence against the null hypothesis of no cointegration, implying a closer long-term relationship between non-wash trade and on-chain activity. The increasing p-values from Quartile 1 to Quartile 4 suggest that at peak wash-trading volumes, on-chain transactions mirror exchange activity the most. Although counter-intuitive—because on-chain activity represents global BTC usage, not solely BTC-USD trades—the pattern suggests that high wash trading may stimulate additional transactions across other platforms and wallets. Thus, periods of elevated wash trading on Mt. Gox may also coincide with increased on-chain activity.

From Table 5, Panel C, I observe that Quartile 4 (high wash-trade periods) also shows the highest coefficient estimate. This mirrors the results from Panels A and B, reinforcing the interpretation that high wash trading corresponds to stronger cointegration and correlation with broader BTC-USD market dynamics. As previously discussed, during low-trading periods, activity on each platform is more idiosyncratic to its specific user base. In contrast, high wash trading may create contagion effects, synchronizing activity

across exchanges.

However, all three panels may suffer from omitted variable bias, as indicated by statistically significant intercept terms. For the on-chain analysis, this could be due to differences between off-chain and on-chain behavior. For the cross-market results, it is plausible that activity in one exchange does not necessarily propagate to others—if it did, trading volumes would be more evenly distributed across exchanges. Yet, Mt. Gox itself controlled approximately 70% of BTC trading volume during the sample period.

3.3 Are Wash Trades Motivated by Other Asset Classes?

The activity of other assets could also serve as leading or lagging factors of wash trades. My reasoning is that when the volatility of other assets decreases, wash traders may use this lull to initiate wash trades in order to stimulate speculation. To determine whether BTC wash trades are correlated with the movement of other assets, I use high-frequency data from the following sources: (1) the S&P 500 Index, (2) Gold Spot Price, (3) the VIX Index, and (4) the EUR-USD exchange rate.

I first convert the timestamps of each dataset to UTC to match the Mt. Gox data. Since the high-frequency data are recorded in 1-minute intervals, I aggregate them into 30-minute intervals, then compute the Amihud illiquidity measure and realized volatility where applicable. Because these markets are not open 24 hours, I impute zero values for intervals when the markets are closed. I then estimate a VAR model with these variables and $wash_t$ as the dependent variable. The impulse response function (IRF) results are presented in Table 6.

In Table 6, we observe that generally, when other asset prices or trading volumes increase, $wash_t$ tends to decrease. This pattern is consistent across the time horizon from $t + 1$ to $t + 10$. These results provide evidence that wash trades are not random and are less likely to occur when other asset markets are more active. This may offer investigators a novel analytical lens: if wash trading aims to amplify BTC activity, it would be strategically inefficient to engage in wash trading when market attention is diverted to other major assets. In such periods, speculative attention on BTC could be

cannibalized by alternatives. Thus, wash trades appear to be strategically timed.

3.4 Does Media Attention Affect Wash Trades' Effects?

In this section, I argue that due to the underdeveloped information environment surrounding cryptocurrencies, wash trades can exert a greater influence on trading activity than in traditional equity markets. Unlike the latter—which benefit from robust and mature information channels including analysts, regulators, institutional disclosures, and corporate governance—cryptocurrencies exist in an information environment that is highly fragmented and often rumor-driven (see Asquith, Mikhail, and Au, 2005; Babenko and Sen, 2015). Therefore, I hypothesize that wash trades are more impactful when the digital asset environment is opaque and public information is scarce.

To empirically examine this, I use Google Trends data on Bitcoin-related searches as a proxy for media attention. I construct a binary variable indicating whether the search interest score for Bitcoin is above or below its median level. I then estimate a VAR model incorporating this media attention variable and present the results in Table 7.

From Table 7, I observe that when media attention is high, non-wash trades respond more rapidly to wash trades. Specifically, in Column (6), the response at $t + 1$ is more pronounced compared to Column (2). However, the cumulative effects in Columns (1)–(2) are greater than in Columns (5)–(6), suggesting that while high media attention accelerates the impact of wash trades on non-wash trades, the effects also dissipate more quickly. This indicates that wash traders may be capitalizing on periods of heightened media interest to execute trades with quicker but less durable influence. Overall, the evidence suggests that wash trades are more potent—albeit shorter-lived—when information asymmetry is greater and rumors dominate the discourse.

3.5 Exogenous Shock: Pot Day, 20 April 2012

Lastly, to determine whether wash trading is strategically motivated by actual demand for Bitcoin, I study an exogenous shock—Pot Day, 20 April 2012. This day was associated with large-scale cannabis sales on the Silk Road, a prominent darknet marketplace at the

time. According to Christin [2013], there was a substantial promotional campaign on Silk Road surrounding Pot Day, with a significant influx of cannabis sellers entering the marketplace approximately two weeks prior to the date. The majority of these transactions were conducted using Bitcoin.

I hypothesize that wash traders may have responded to this exogenous shock by increasing their wash trade activity in advance of Pot Day to capitalize on the anticipated surge in legitimate BTC demand. If so, this would imply that wash trading is not only responsive to market dynamics but also strategically timed around external demand-driven events.

From Table 8, by observing the sum of the impulse response functions (IRFs), we see a notable drop in the cumulative IRF values of both $total_t$ and $nonwash_t$ between Columns (1)–(2) and Columns (3)–(4): from 7.54×10^4 to 4.63×10^4. This indicates that trading activity was less responsive to wash trades immediately following Pot Day. The evidence suggests that the effectiveness of wash trades is diminished in the presence of strong contemporaneous events that independently drive BTC demand.

Furthermore, when non-wash trade activity increased, Columns (4) and (8) both show a general decrease in wash trading, consistent with the prediction that in more active markets, wash traders pivot toward engaging with genuine buyers rather than simulating activity. The comparison of IRF magnitudes—-8.61×10^3 in Column (4) versus -6.94×10^4 in Column (8)—also indicates that wash traders reduced their activity to a lesser extent after Pot Day when non-wash trade volumes rose. This suggests that they likely intensified wash trading before Pot Day to generate hype and subsequently scaled back afterward.

Overall, these findings support the interpretation that wash trading behavior was strategic during this period, aligned with a contemporaneous demand shock, and heightened during periods when Bitcoin demand was expected to spike.

4 Conclusion

The results in this paper show that wash-trading is indeed strategic and do not occur in a vacuum. I document certain strategic factors that may be of interest to federal investigators and other cryptocurrency stakeholders who may be interested in investigating wash-trading activity.

Firstly, lagged non-wash trade volume and volatility triggers more wash trading, showing that wash trading takes place strategically. The results are robust to a placebo test. Secondly, when wash-trading in an exchange is high, the exchange's activity mirrors the on-chain transactions and other markets' transactions. This is most probably because when trading is low, the on-chain transactions and activity in other markets are more idiosyncratic to their own customers and more general. When wash trading is high, the contagion effect causes every one's activity to be more similar.

Thirdly, wash-trading also takes place more when trading on other important asset classes are lower in volume. Fourthly, when online rumor interest is high, wash trades will cause the volume traded to go up fast but also recedes more quickly than if there is no media interest. In other words, when the information environment is weaker such as during lower google search interest time periods, other legitimate cryptocurrency transactions could be fueled by information asymmetry according to traditional wash trading theory and the response is slower but longer-lasting. Finally, when there are contemporary demand events that would likely trigger more customers to buy bitcoin, wash trading increases more to take advantage of this speculation.

One limitation of this article is that by using the Mt.Gox dataset, there is a tradeoff between verifiability and timeliness. Current wash trading activity may differ from the past during Mt.Gox's time due to the different macroeconomic factors surrounding BTC currently.

Future research can extend the techniques developed in this study by applying them to more recent or proprietary transaction data from cryptocurrency exchanges beyond the Mt. Gox dataset. While existing literature often relies on general statistical patterns—such as Zipf's Law, Benford's Law, and trade-size clustering—to provide *prima*

facie evidence of wash trading at the exchange level, more granular and robust methods are possible when detailed transactional data are available. In particular, machine learning approaches such as Classification and Regression Trees (CART) and Long Short-Term Memory (LSTM) models offer promise in profiling wash trading behavior at the trade level. These tools can support both market surveillance efforts and legal enforcement actions aimed at detecting and prosecuting wash trading.

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